Name:_____

# Newton & His Laws

Force= a push or a pull on an object.  It is measured in newtons (N).

**Newtons first law of Motion=**

A tendency of an object to _____ a change in its motion.  Also called i_____.

-an object that is moving will continue to move at the same _____ and in the same _____ unless an _____ force acts upon it.

**Newtons second law of motion=**

Acceleration is produced when a force acts on a mass. The greater the m_____ (of the object being accelerated) the greater the amount of f_____ needed (to accelerate the object).

F = _____ and is measured in _____

M= _____ and is measured in _____

A= _____ and is measured in _____

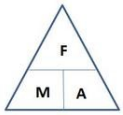
Example

What force is needed to make a 7kg rock accelerates at 2.5 m/s?

F:_____

M: _____

A: _____




F = m x a

F =

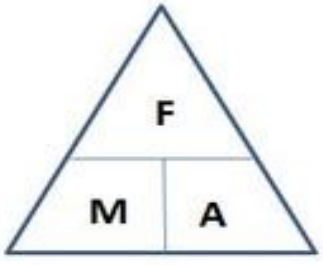
= N

A 10 kg ball is thrown with a force of 20 N.  What will its acceleration be?

F:_____

M: _____

A=f/m



A: _____ /

= $m/s^2$

Questions

1. A force of 100N is applied to a ball of mass 2kg. What will the acceleration be?

2. A 21kg steel ball is accelerating at $3m/s^2$. What force was needed to bring this about?

3. What force will be needed to make a 2kg rock accelerate at $10m/s^2$?

4. A force of 10N is applied to a toy car of mass 1kg. What will the acceleration be?

5. A 200kg car has a force of 4000N applied to it. What will the acceleration be?

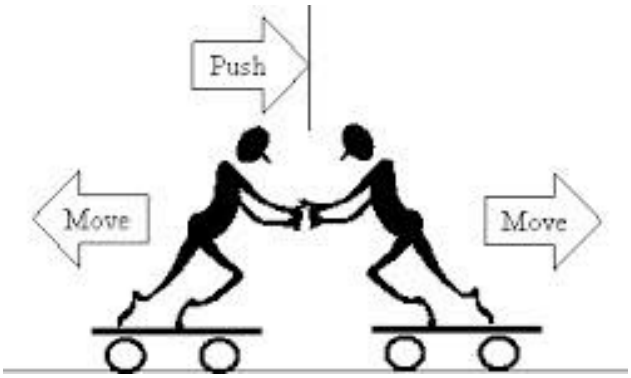
6. A car accelerates at 5m/s. It required a force of 5000N to do accelerate at this rate. What is its mass?

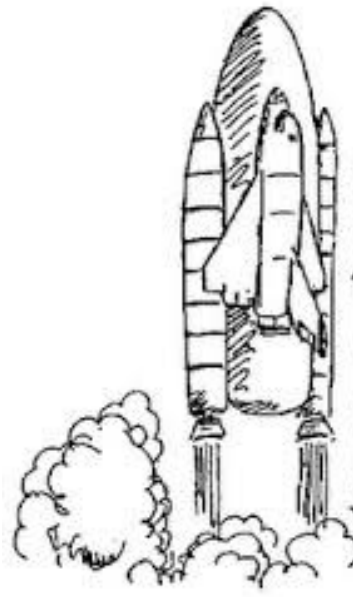
**Newtons third Law of Motion=**

For every _____ there is an _____ and_____ reaction.

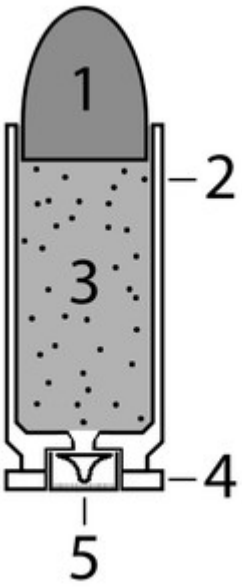
Give an example where you have experienced this: _____
_____
_____

Explain what is going on in the diagrams below:



Bob hits a live bullet with a hammer.  The bullet explodes.  The lead projectile (part 1) flies forward and the brass casing of the bullet (parts 2 and 4) fly backwards and hits Bob in the hand.




Explain what has happened here according to newtons 3 law of motion.